# An exact separator, with source limits

Observed: at auxiliary prime 199, two predicates retain all 10 publications and reject all 11 omissions across six streams. Historical use remains unestablished.

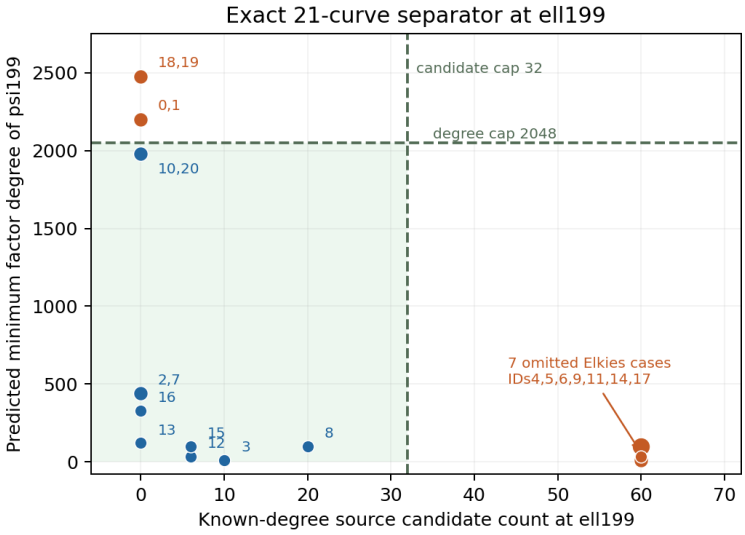
## Known-degree candidate-list size <=32

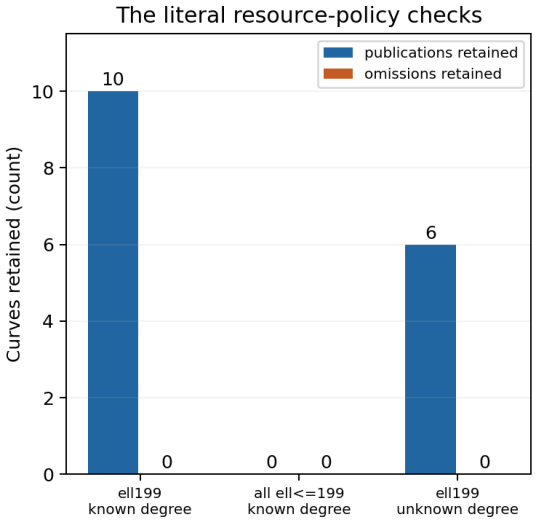
## AND minimum psi199 factor degree <=2048

Posthoc representative caps; the exact constants are not uniquely identified.

Seven omitted Elkies cases have 60 candidates; the four omitted Atkin cases have predicted minimum factor degrees 2200 or 2475. Published Elkies cases have at most 20 candidates; published Atkin cases have degree at most 1980.

The prior library found no two-atom fit. The signed-action extension adds 6680 predicates and now yields 16 fitting signature pairs, 19 truth tables. Both independent arithmetic and complete truth-table implementations agree. All fits are retained.




NIST-SIGNED-FROBENIUS-01 + NIST-ELL199-POLICY-01. Exact fixed N=21; no sampling uncertainty.
Left labels are row indices in ell199.csv. Caps32/2048 are posthoc choices; factor degrees are derived, not full-cohort polynomial factorizations.

The same caps applied uniformly through 199 reject every publication. Unknown splitting-degree mode at 199 rejects four publications. The later counter replay stopped below 199 for all eleven omissions. These are direct limits on a resource-policy explanation.

# Every curve at auxiliary prime 199

P=published; O=omitted. S=split/Elkies; A=nonsplit/Atkin. R is projective order; F is full signed order; C is known-degree candidate count; D is minimum predicted division-polynomial factor degree.

| ID | Curve / ordinal | Type | R | F | C | D |
|---|---|---|---|---|---|---|
| 0 | O skip-863711343-36 | A | 200 | 4400 | 0 | 2200 |
| 1 | O skip-863711343-1562 | A | 200 | 4400 | 0 | 2200 |
| 2 | P prime239v1 | A | 40 | 880 | 0 | 440 |
| 3 | P prime239v2 | S | 22 | 22 | 10 | 11 |
| 4 | O skip-863711343-2491 | S | 198 | 198 | 60 | 99 |
| 5 | O skip-863711343-2682 | S | 198 | 198 | 60 | 99 |
| 6 | O skip-863711343-2863 | S | 198 | 198 | 60 | 99 |
| 7 | P prime239v3 | A | 40 | 880 | 0 | 440 |
| 8 | P prime192v1 | S | 33 | 198 | 20 | 99 |
| 9 | O skip-863966734-2198 | S | 99 | 99 | 60 | 11 |
| 10 | P prime192v3 | A | 20 | 3960 | 0 | 1980 |
| 11 | O skip-863966734-5405 | S | 99 | 99 | 60 | 33 |
| 12 | P prime192v2 | S | 9 | 99 | 6 | 33 |
| 13 | P secp521r1 | A | 40 | 240 | 0 | 120 |
| 14 | O skip-871696354-876 | S | 99 | 99 | 60 | 99 |
| 15 | P prime256v1 | S | 9 | 99 | 6 | 99 |
| 16 | P secp384r1 | A | 10 | 660 | 0 | 330 |
| 17 | O skip-891607927-777 | S | 99 | 99 | 60 | 11 |
| 18 | O skip-891607927-1485 | A | 25 | 4950 | 0 | 2475 |
| 19 | O skip-891607927-5736 | A | 25 | 4950 | 0 | 2475 |
| 20 | P secp224r1 | A | 20 | 3960 | 0 | 1980 |

Exact field/model/order/seed identities and full UIDs are in prior-rows.json; all 199 values are in ell199.csv. Nonsplit cases have C=0 because that source eigenvalue list is skipped. D is derived from the signed action, not newly factored here.

Every candidate cap 20..59 paired with every degree cap 1980..2199 gives the same observed split. 32/2048 are convenient examples, not recovered constants. No omitted position is asserted to have been visited by the original program.

# The policy checks and source obstructions

At 199, known degree: 10 publications retained; 0 omissions retained.

Uniform through 199, known degree: 0 publications retained; 0 omissions retained.

At 199, unknown degree: 6 publications retained; 0 omissions retained.

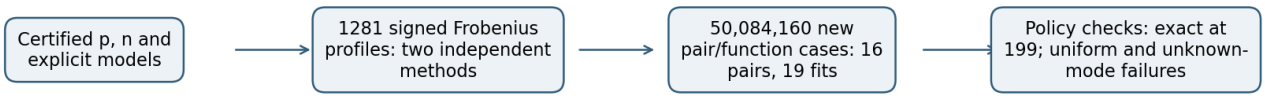
Unknown-degree mode loses prime239v2, prime192v1, prime192v2 andprime256v1. The uniform caps fail every publication, first at auxiliary primes 67 through 83. The original degree mode and 199 requirement are unestablished.

In the later native-counter observations, all 11 omissions stopped before 199. Only the P-384/P-521 controls reached 199 or later; published P-224 stopped at 197. An at 199 callback cannot explain those omission decisions in that replay. This does not establish original 1997 visitation.

A list-length cap is different from a comparison timeout. Omitted 239/2863 and 256/876 have 60 candidates but correct positive-half ranks 3 or 8. A32-position search limit would admit either. The recovered source allocates 100 entries at 199, not 32.

The smallest irreducible factor is different from every intermediate polynomial. The period conjugate construction can have degree r*(ell-1)/2. Published 239v1,239v3 and P-521 have r199=40, giving degree 3960, even though their minimum irreducible degrees are 440,440,120. A hard2048 bound on all intermediate polynomials fails them.

No recovered source prescribes the numerical caps or a mandatory prime 199 check. The exact classifier is retained as a specific lead. Actual selection criteria remain unidentified.




Original caller, caps and199 requirement remain unestablished.

# Validation, torsion meaning and provenance

All 1281 cells agree independently: 673 nonsplit, 574 split, 33 non-scalar Jordan, 1 scalar. Direct quotient coordinate maps settle the three ambiguous scalarity cells. Predicted division-polynomial factor degrees agree by separate matrix fixed-point and eigenvalue-cycle formulas.

P-224 at 3 is the scalar -I cell: full Frobenius order 2, order modulo sign 1, projective order 1. Its quadratic twist therefore has full rational 3-torsion. The three P-224 omissions have Jordan action at 3. This explains a concrete torsion aspect of the cube/splitting pattern, not why the original program selected it.

Controls +I,-I, Jordan, nonsplit give full/x/projective orders(1,1,1),(2,1,1),(3,3,3),(48,24,8). Native counts and division-polynomial factors, independent GP control counts, A/A and invalid-model/order controls pass. Local LMFDB class labels1.7.b, 1.7.ab, 1.7.e, 1.5.d are class provenance only; models are independently counted.

The new 1313 columns and 6680 atoms give 2056 new signatures. 3,130,260 new pair classes and 50,084,160 truth tables were exhaustively checked twice. 16 pairs support 19 fitting tables; no new single predicate fits. Prior old-old zero-fit pairs are reused. All aliases, fits and nonfit witnesses remain available; all labels were already visible.

Primary sources: Schoof 1995, doi 10.5802/jtnb.142; Morain 1995, doi 10.5802/jtnb.143. The latter cites Mueller/Dewaghe 1994 work in preparation. This does not identify those manuscripts with a later routine or establish original caller use. Source code is later LiDIA 23dac491, explicitly unproven as the 1997 implementation.

Both formal exports pass file verification. Archive: 54 sealed files, SHA256 e8c128fc7f78497775b83f2e8e49cbe6b92da9a12b7c6993ef293425619bd388. Original paper/package bytes are excluded. Canonical upload returned HeadObject403; exact bytes remain in Git under the existing PR2602 exception.

Relocation validation hashes all payloads and replays both 1281-cell methods, all bitset pairs and the three policies. It compares with the original complete rowwise results/witnesses; rowwise enumeration is not repeated. Replays are same-data validation. See PORTABLE-REPLAY.json, PLAN.md, ALL-FITS.json and both protocols.

Next discriminating evidence: an authenticated requirement/caller or source-backed reason for the 199 gate, degree mode and actual limits. The selector problem and goal remain open.